

LoRA Privacy Leakage — Identifiability Results

How much of the private fine-tuning data the LoRA adapter makes recoverable — the rank sweep and its mechanism.
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Overview — what is measured, and what is (and isn't) shown

What this shows. Fine-tuning a small image classifier with LoRA on a few PRIVATE images leaves a recoverable trace in the released adapter (the A, B matrices). This deck reports the SOLID, verified result — IDENTIFIABILITY:

q_{eff} = how many independent directions of the private data the adapter makes recoverable IN PRINCIPLE — an information / capacity number (a Fisher-rank), stated as “ $\geq X$ recoverable directions” (a conservative lower bound). It measures whether — and how much of — the secret is recoverable, not the pixels.

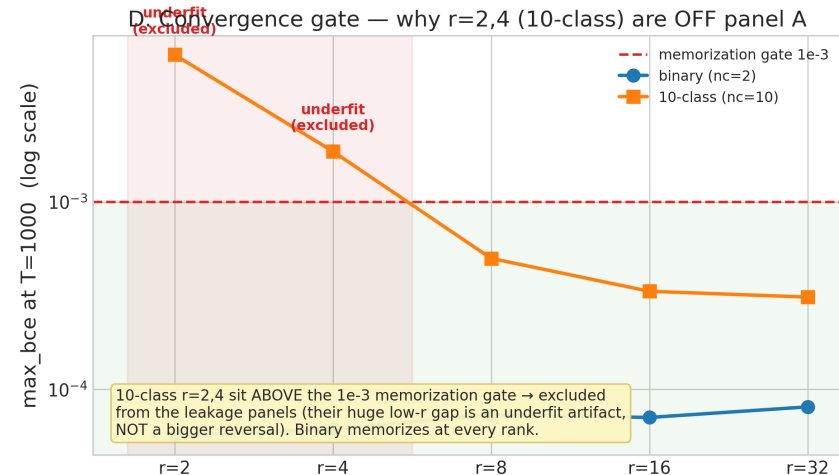
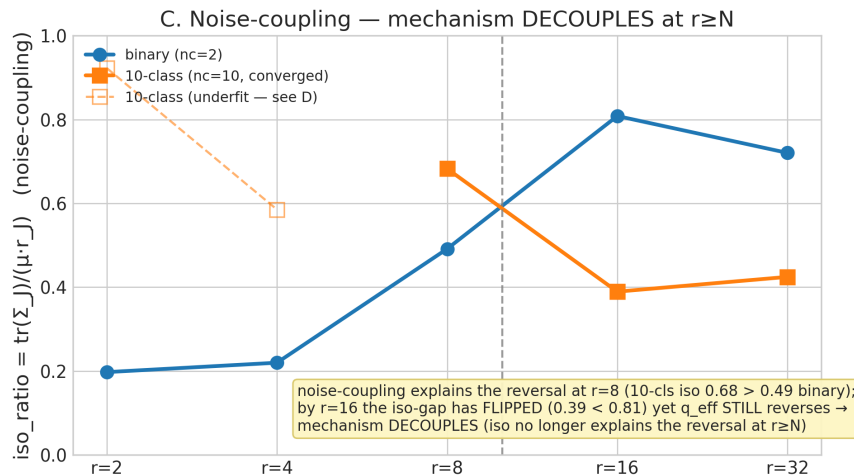
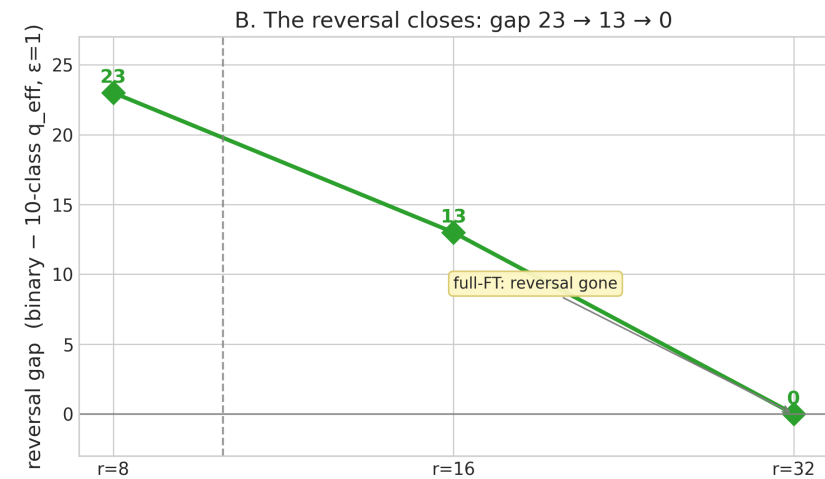
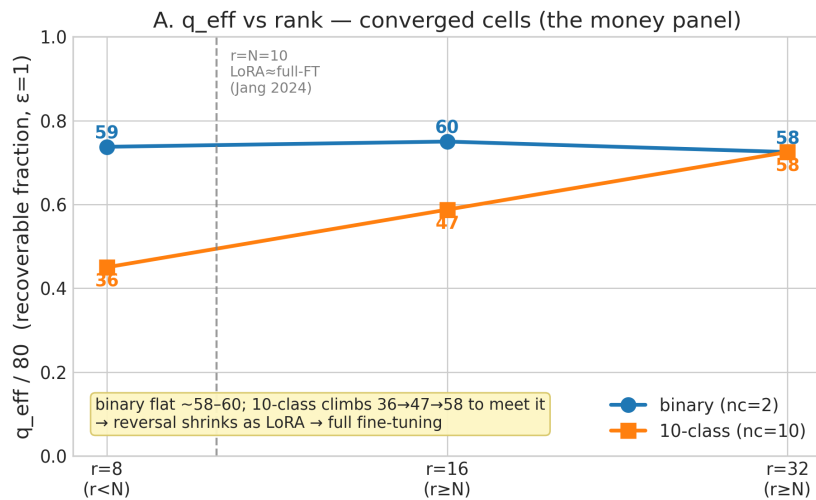
Headline result. Multi-class cross-entropy does NOT amplify leakage. At convergence it leaks FEWER directions than binary (a “reversal”), and that reversal is a genuinely LOW-RANK effect that vanishes as LoRA approaches full fine-tuning (rank $\geq$ the number of private images, $N=10$).

Not shown (deliberately). Turning these recoverable directions into actual PIXELS — the gradient-bridge reconstruction track — does NOT yet work as a result. Independent check across all 20 reconstruction result files: the adapter-only reconstruction fails to beat the trivial mean-image baseline in 0 of 40 cases (raw SSIM 0.22-0.44 vs a mean-predictor's 0.56-0.76 on MNIST) — i.e. no demonstrated instance-specific information. The SSIM $\approx$ 0.57-0.61 quoted elsewhere is a mean/std-matched (inflated) metric that was never compared to that baseline. So pixel reconstruction is an OPEN LIMITATION, not a result, and is omitted here; closing it is what the generative-prior direction is for.

The reversal is a LOW-RANK effect

How to read: x-axis = LoRA rank (log-scale 2-32). The money panel plots recoverable directions q_{eff} vs rank for binary vs 10-class, CONVERGED ranks only. The reversal gap (binary – 10-class) shrinks 23 → 13 → 0 as rank crosses $N=10$ (the vertical band) and reaches ZERO at $r=32$, where LoRA $\approx$ full fine-tuning (Jang 2024). The separate convergence-gate panel shows WHY $r=2,4$ are excluded from the money panel: the 10-class base fails to memorize there (max_bce above the $1e-3$ line), so those points are convergence-confounded, not real leakage. Takeaway: the multi-class privacy advantage is specific to low-rank LoRA and disappears at full-fine-tuning rank. Anchor $r=8$ reproduces the locked run exactly (q_{eff} 59/36, iso 0.49/0.68); $r_J=80$ at every rank.

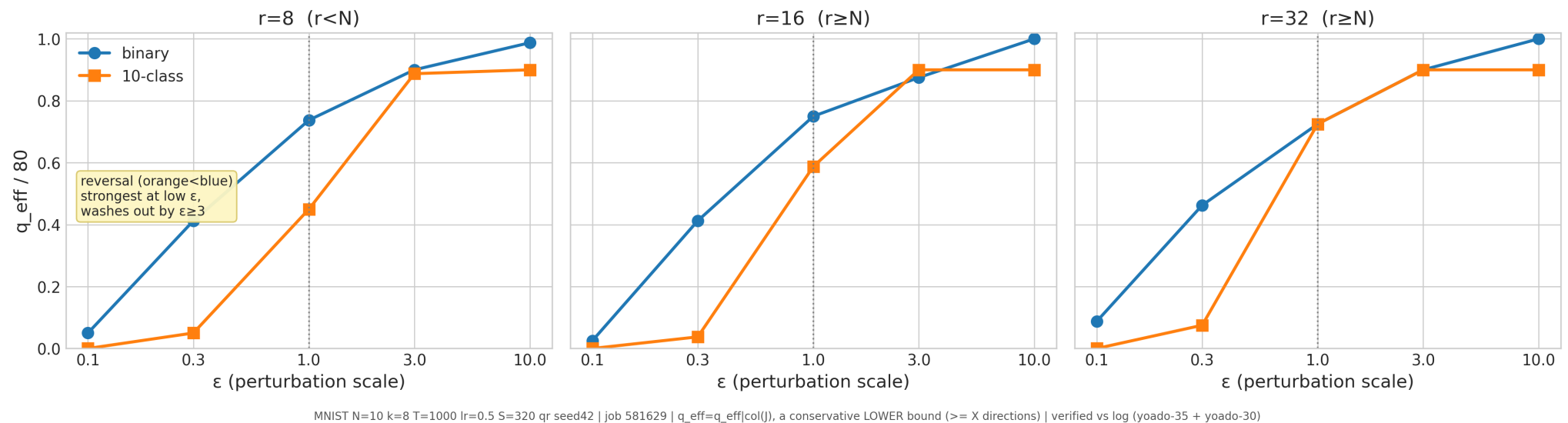
LoRA rank sweep — the multi-class q_{eff} reversal is a LOW-RANK effect that vanishes at full fine-tuning ($r \geq N$)



ϵ -dependence — the headline is not cherry-picked

How to read: q_{eff} vs the perturbation scale ϵ (log-x), one small panel per converged rank, both bases. The reversal (10-class line BELOW binary) is clearest at small ϵ — the noise-limited regime — and washes out by $\epsilon \geq 3$, where strong perturbations overwhelm the noise floor. This confirms the $\epsilon=1$ slice used in the headline is representative, not a cherry-pick, and it sits exactly where the noise-floor mechanism predicts.

q_{eff} vs ϵ per converged rank — the reversal lives at low ϵ ($\epsilon=1$ in Fig 1 is representative, not cherry-picked)



The geometry behind the count (r=8)

How to read: the sorted whitened signal-to-noise spectrum $\sigma_i(j_SNR)$ at rank 8, both bases, with the $\varepsilon=1$ recovery threshold marked. q_eff literally COUNTS the directions whose SNR clears that line. 10-class pushes more directions BELOW the noise floor, so it recovers fewer — this is the geometric picture behind the number. (Caveat, stated honestly on the figure: this noise-coupling mechanism explains the reversal at $r=8$, but DECOUPLES at $r \geq N$ — the reversal there persists against the mechanism's prediction, so the WHY is settled only at low rank.)

Geometry behind the count (r=8) — 10-class has more directions below the noise floor. NOT a reconstruction.

